ANKARA UNIVERSITY
FACULTY OF ENGINEERING
DEPARTMENT OF COMPUTER ENGINEERING

COM4569/ CMP4569 DATA MINING
2026-2027 FALL
TERM PROJECT GUIDELINE

This guideline describes the preparation process and assessment criteria for the Data Mining term project.

Team size and submission: Project teams must consist of four or five students.

Each report must be prepared as a single .docx or .pdf file and uploaded to the designated assignment submission area on the course's e-Kampüs page. A deduction of 10% of the total possible score will apply for each day of delay. Late submissions will be accepted for a maximum of two days after the deadline.

Grading: The project accounts for 20% of the overall course grade: 5% for the project proposal, 5% for the presentation and demo, and 10% for the final report.

Project Proposal (5%): Submission deadline: October 25, 2026 (23:59).

Use the template provided on the course page. Specify the project topic and team members' names. State the project objective and provide a brief summary. Cite previous studies related to your topic. Describe the dataset, the project tasks, and each team member's responsibilities.

Final project report (10%): Submission deadline: December 13, 2026 (23:59).

Prepare the final report using the sections and assessment criteria on the following page. The report is graded out of 100 points.

Project Presentation and Demo (5%): December 17- December 24 during lecture hours

Each team will have 15 minutes to present its project. During the demo, run the models developed for data mining tasks, such as classification, in WEKA, Orange, or a similar tool. Explain the project by comparing the accuracy scores obtained. All team members must participate in the presentation.

Final Report Content and Assessment

Section 1. Introduction and Related Work 10 points

Explain the objective and significance of your project. Present at least two relevant studies from the literature; you may use the sources cited in your proposal. Summarize each study in one paragraph, covering the data used, preprocessing methods, data mining techniques, and results, such as accuracy scores. Provide the corresponding references at the end of the report.

Section 2. Problem Definition and Data Mining Algorithms 20 points

Task definition: Describe the problem and approach. State whether the task involves supervised or unsupervised learning, such as classification or clustering.

Selected algorithms: Choose at least three data mining algorithms for comparison. You may include brief descriptions of the selected algorithms in this section.

Section 3. Data Overview 10 points

Describe the dataset, including its source, features, and class attribute, where applicable.

Section 4. Data Preparation 20 points

Data preprocessing and cleaning: Does your dataset contain missing values, outliers, or other issues? Explain how these issues can be detected.

Feature selection: Describe the feature selection step and identify the algorithms you used, with implementations in WEKA, Orange, or Python.

Section 5. Results 20 points

Describe model selection and training. Compare the accuracy scores of the selected algorithms. You may present a table comparing results obtained using all features with those obtained using selected features.

Section 6. Discussion and Conclusion 10 points

Significance and contributions: Discuss which data mining algorithm is most suitable for your project. Compare your findings with previous studies.

Limitations and future work: Explain any problems or limitations encountered and suggest ways to extend the work.

References

Use the IEEE numbered citation style, for example [1], [2], and so on.

Final report format 10 points

Include a cover page stating the project title and team members. Organize each step into a separate section. Use 2.5 cm margins on all sides, 12-point Times New Roman, 1.5 line spacing, and justified alignment. List all cited sources in the references section at the end of the report.